Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
aagibadullin@aagibadullin:~$ ls /etc/ > file.txt
aagibadullin@aagibadullin:~$ ls >> file.txt
aagibadullin@aagibadullin:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
aagibadullin@aagibadullin:~$ grep .conf file.txt > conf.txt
aagibadullin@aagibadullin:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage04/report/_resources/csl
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage04/report/bib/cite.bib
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage05/report/_resources/csl
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage05/report/bib/cite.bib
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage06/report/_resources/csl
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
oject-personal/stage06/report/bib/cite.bib
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
esentation/report/_resources/csl
/home/aagibadullin/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/pr
esentation/report/bib/cite.bib
/home/aagibadullin/git-extended/.git/hooks/commit-msg.sample
/home/aagibadullin/git-extended/.git/objects/ce
/home/aagibadullin/git-extended/.git/objects/c5
/home/aagibadullin/git-extended/.git/objects/d7/cdfff8514284f6653890e17e6c1de8103acba3
/home/aagibadullin/git-extended/.git/config
/home/aagibadullin/conf.txt
aagibadullin@aagibadullin:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/foomatic/hashes.d/hashes.new
find: ‘/etc/libvirt’: Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
aagibadullin@aagibadullin:~$
aagibadullin@aagibadullin:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
aagibadullin@aagibadullin:~$
aagibadullin@aagibadullin:~$
aagibadullin@aagibadullin:~$ find ~ -name "log*" > logfile &
[1] 6260
aagibadullin@aagibadullin:~$
[1]+  Done                    find ~ -name "log*" > logfile
aagibadullin@aagibadullin:~$ rm logfile
aagibadullin@aagibadullin:~$ kill 6260
bash: kill: (6260) - No such process
aagibadullin@aagibadullin:~$
```

Рисунок 5: Фоновый запуск процесса

```
aagibadullin@aagibadullin:~$
aagibadullin@aagibadullin:~$ gedit &
[1] 6308
aagibadullin@aagibadullin:~$ ps | grep gedit
   6308 pts/1    00:00:00 gedit
aagibadullin@aagibadullin:~$ kill 6308
[1]+  Terminated              gedit
aagibadullin@aagibadullin:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                              User Commands                              DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents the GNU version of df.  df displays the amount of
       space available on the file system containing each file name argument.   If  no
       file  name  is given, the space available on all currently mounted file systems
       is shown.  Space is shown in 1K blocks by default, unless the environment vari-
       able POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node containing a  mounted
       file  system,  df  shows the space available on that file system rather than on
       the file system containing the device node.  This version of df cannot show the
       space available on unmounted file systems, because on most kinds of systems do-
       ing so requires non-portable intimate knowledge of file system structures.

OPTIONS
       Show information about the file system on which each FILE resides, or all  file
       systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

       -B, --block-size=SIZE
 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                                User Commands                                DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes  rather  than device usage; although the apparent
              size is usually smaller, it may be larger due  to  holes  in  ('sparse')
              files, internal fragmentation, indirect blocks, and the like

       -B, --block-size=SIZE
              scale  sizes  by  SIZE before printing them; e.g., '-BM' prints sizes in
              units of 1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
aagibadullin@aagibadullin:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89637972  64880956  59% /
devtmpfs         4011288        0   4011288   0% /dev
tmpfs            4035664       92   4035572   1% /dev/shm
tmpfs            1614268     1952   1612316   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journald.serv
ice
/dev/nvme0n1p3 155186176 89637972  64880956  59% /home
tmpfs            4035664     1612   4034052   1% /tmp
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop0        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
/dev/loop1         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolved.serv
ice
tmpfs             807132      172    806960   1% /run/user/1026
/dev/sr0         2677920  2677920         0 100% /run/media/aagibadullin/Fedora-WS-Live
-43
aagibadullin@aagibadullin:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4       ./git-extended/.git/objects/d7
4       ./git-extended/.git/objects/dd
4       ./git-extended/.git/objects/1b
4       ./git-extended/.git/objects/47
4       ./git-extended/.git/objects/a1
76      ./git-extended/.git/objects
8       ./git-extended/.git/logs/refs/heads
8       ./git-extended/.git/logs/refs/remotes/origin
8       ./git-extended/.git/logs/refs/remotes
16      ./git-extended/.git/logs/refs
20      ./git-extended/.git/logs
212     ./git-extended/.git
220     ./git-extended
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
848456  .
aagibadullin@aagibadullin:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/aagibadullin/git-extended/.git/objects/75
/home/aagibadullin/git-extended/.git/objects/2d
/home/aagibadullin/git-extended/.git/objects/68
/home/aagibadullin/git-extended/.git/objects/b0
/home/aagibadullin/git-extended/.git/objects/04
/home/aagibadullin/git-extended/.git/objects/ec
/home/aagibadullin/git-extended/.git/objects/d7
/home/aagibadullin/git-extended/.git/objects/dd
/home/aagibadullin/git-extended/.git/objects/1b
/home/aagibadullin/git-extended/.git/objects/47
/home/aagibadullin/git-extended/.git/objects/a1
/home/aagibadullin/git-extended/.git/logs
/home/aagibadullin/git-extended/.git/logs/refs
/home/aagibadullin/git-extended/.git/logs/refs/heads
/home/aagibadullin/git-extended/.git/logs/refs/remotes
/home/aagibadullin/git-extended/.git/logs/refs/remotes/origin
/home/aagibadullin/monthly
/home/aagibadullin/reports
/home/aagibadullin/reports/monthly
/home/aagibadullin/reports/monthly/monthly
/home/aagibadullin/ski.plases
/home/aagibadullin/ski.plases/equipment
/home/aagibadullin/ski.plases/plans
/home/aagibadullin/australia
/home/aagibadullin/play
/home/aagibadullin/play/games
/home/aagibadullin/play/games/play
aagibadullin@aagibadullin:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.